Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/11/2018
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G, with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism is an International Space Station Experiment to investigate the effects of gravity on: 1.) electrically generated two-phase flow and 2.) electrically driven liquid film boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of		

Task Description:	this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This is a proposal to support the development of the experiment flight hardware for the International Space Station (ISS), conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. The robust collaboration between Worcester Polytechnic Institute and Goddard Space Flight Center can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices.
Task Progress:	Science Review Document: The Science Review Document (SRD) was revised and resubmitted by the investigators per the comments received from Dr. M.M. Hasan of NASA Glenn Research Center throughout the above mentioned period. The revisions addressed the comments that were raised by the SRD panel. Fabrication of the EHD Pumping Plate and Heaters: We investigated two manufacturing processes for EHD pumping plate and heaters to characterize the limitations associated with each process: E-beam and sputtering. In each case, materials and pump dimensions were found satisfactory to experiment goals.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Peer-reviewed Journals	Yang L, Talmor M, Shaw BC, Minchev KS, Jiang C, Seyed-Yagoobi J. "Flow distribution control in meso scale via electrohydrodynamic conduction pumping." IEEE Transactions on Industry Applications. 2017 Mar-Apr;53(2):1431-8. http://dx.doi.org/10.1109/TIA.2016.2626358 , Mar-2017
Papers from Meeting Proceedings	Yang L, Barruetabena J, McGuirk CD, Ouellette LM, Yagoobi J. "Experimental and Numerical Study of Fluid Thermal Homogenization in a Spherical Reservoir with EHD Conduction Pumping Mechanism." 2017 IEEE 19th International Conference on Dielectric Liquids (ICDL), Manchester, England, June 25-29, 2017. In: 2017 IEEE 19th International Conference on Dielectric Liquids (ICDL), Manchester, England, June 25-29, 2017. 4 p. https://doi.org/10.1109/ICDL.2017.8124596 , Jun-2017
Papers from Meeting Proceedings	Yang L, Yagoobi J. "Liquid-Phase Flow Distribution Control in Meso-Scale with Directionally Reversed Electrohydrodynamic Conduction Pumping Configuration." 2017 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 13-15, 2017. In: Proceedings of the 2017 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 13-15, 2017. http://www.electrostatics.org/esa2017proceedings.html ; http://www.electrostatics.org/images/2017_K1.pdf , Jun-2017
Papers from Meeting Proceedings	Talmor M, Lei Y, Yagoobi J. "Numerical Study of Directional EHD Conduction Driven Flow Distribution Control in Small Scales in the Presence of External Pressure Loads." 2017 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 13-15, 2017. In: Proceedings of the 2017 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 13-15, 2017. http://www.electrostatics.org/esa2017proceedings.html ; http://www.electrostatics.org/images/2017_K7.pdf , Jun-2017